



Feedback Control Systems (EE3004)

Sessional-I Exam

Date: February 25th 2025

Total Time (Hrs): 1

Course Instructor

Total Marks: 30

Dr. Omer Saleem & Ms. Tamania Javed

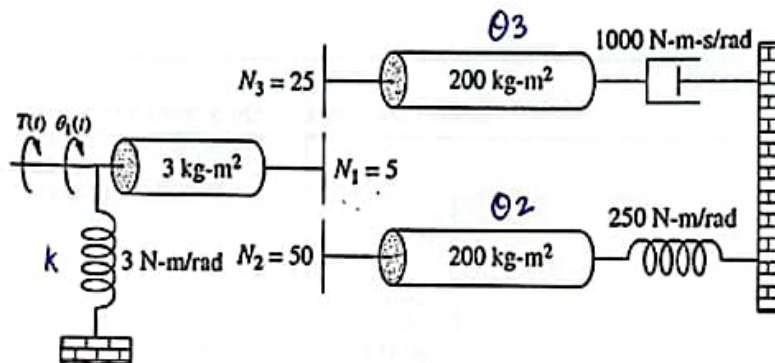
Total Questions: 3

Do not write below this line

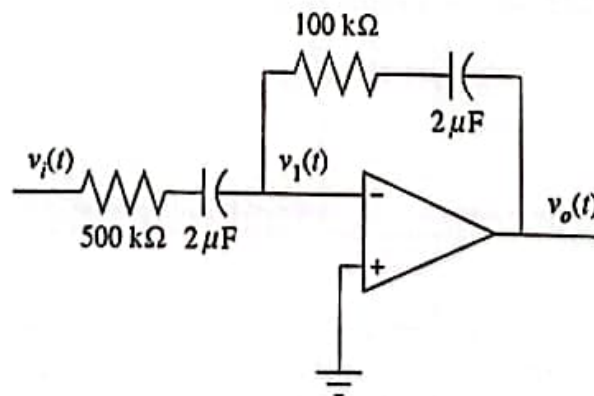
Attempt all the questions.

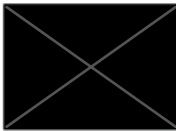
CLO # 1: Develop the mathematical model of a physical system

Q1: Formulate the transfer function, $G(s) = \theta_1(s) / T(s)$, for the rotational mechanical system shown in the Figure below. [10 marks]

**CLO # 1: Develop the mathematical model of a physical system**

Q2: Formulate the transfer function $G(s) = V_o(s) / V_i(s)$ of the following inverting-amplifier circuit. Hence, use this transfer function to evaluate $v_o(t)$, if a unit-step signal is applied at $v_i(t)$. [10 marks]



**CLO # 1: Develop the mathematical model of a physical system**

Q3: Assume that currents $i_1(t)$ and $i_2(t)$ are flowing in loop 1 and loop 2 of the following circuit, respectively. The input of the circuit is $v_i(t)$. The voltage across the capacitor is $v_C(t)$. The output of the circuit is $v_R(t)$. Develop the state-space model of the system, identifying the state variables as well as the matrices A , B , C , and D . [10 marks]

